



Data Driven Education

Discussion Forum Subteam

Presentation 1 - 02/12/2025

Meet the Team



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Introduction

- Online Discussion platform is crucial in modern education.
- **Ed Discussion:** a widely used online discussion forum
 - enables students and instructors to interact
 - discuss course materials asynchronously
- How is Online Discussion Forums different?
- What to do to better utilize Online Discussion platforms?
- Improve the whole education experience

Research Questions

- What discussion themes are most commonly present in the online course forums?
- Can sentiment analysis of discussion posts provide insights into student engagement and attitudes toward the course
- How do different behaviors correlate with student success?
- How can topic modeling (LDA) and classification models (BERT?TF-IDF) improve the automatic categorization of forum posts?
- What is the best way to present discussion analytics to instructors in an actionable and user-friendly dashboard?

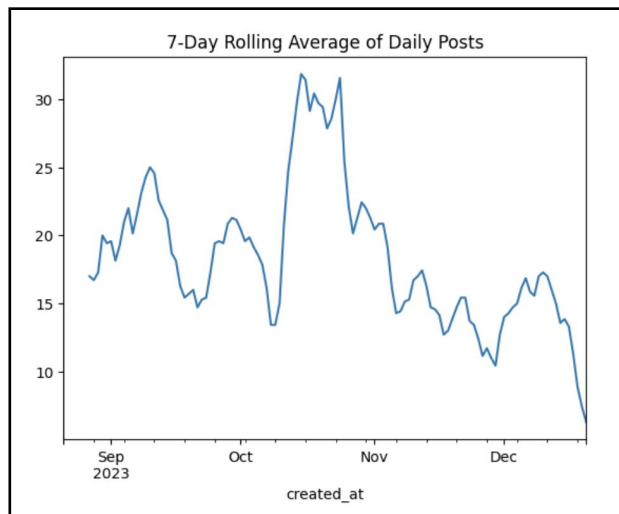
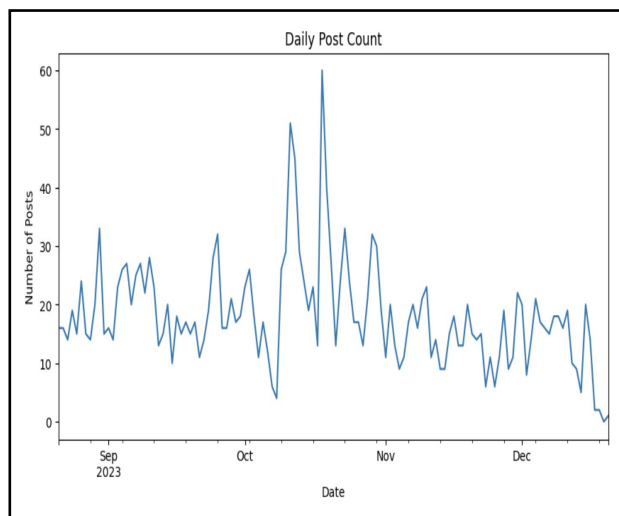
Recap of Fall 2024 Efforts:

Data Science:

- Time series-analysis on dummy-data
- Conducted Entropy & Type-Token Ratio studies
- Insights for engagement fluctuations & student behavior

Machine Learning:

- Developed a preliminary synthetic dataset
- Refined output of generative AI model
- Explored use of integrating datasets to improve output

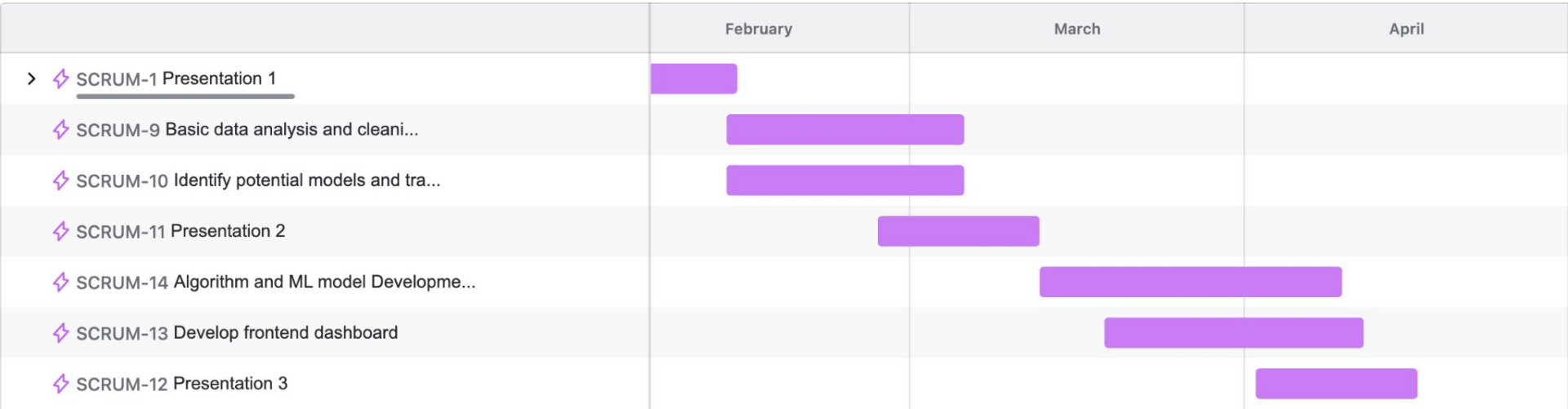


Goals for the Semester

- Enhance analysis of online discussion form data
 - Data-driven insights, ML models, and more
 - Develop a dashboard
- Want to develop engagement metrics
 - Student/instructor participation
- Explore topic modeling (LDA, BERT, etc)
- Analyze sentiment trends
- Explore Data Science efforts w/ related research papers
- Improve output of synthetic dataset generation



Project Timeline

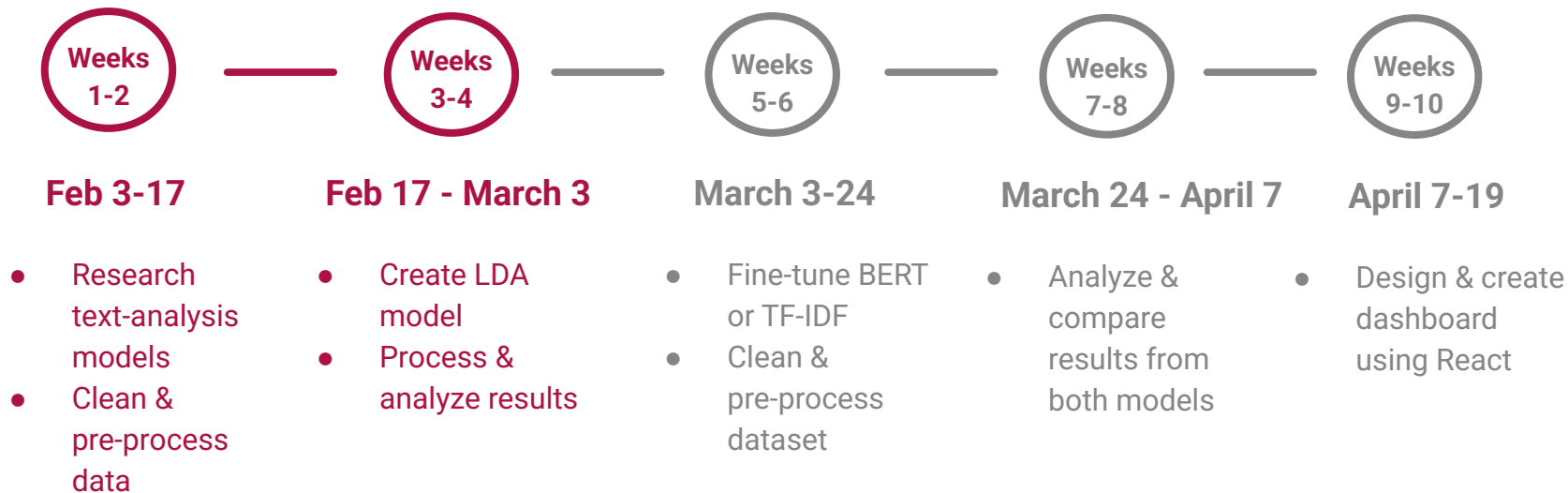


Jira - For task tracking and management

Data Science

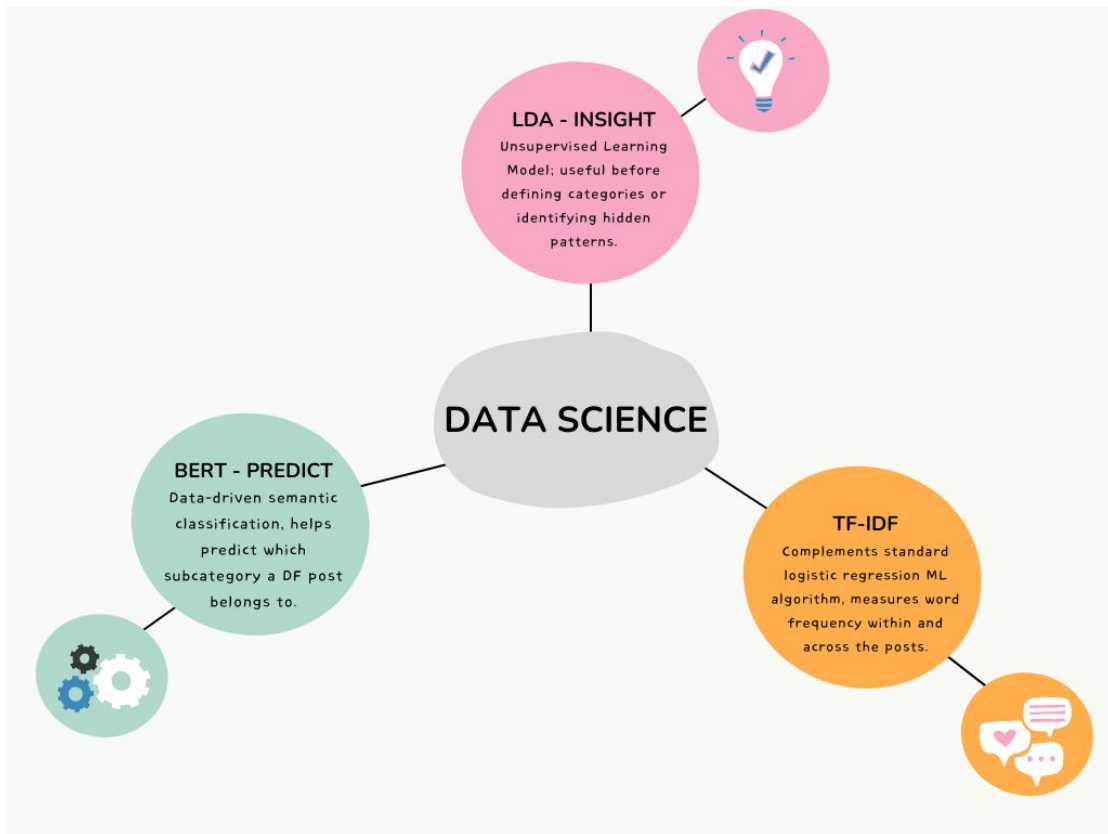
Goals & Timeline for the Semester

- Explore **text-analysis models** for topic modeling & sentiment analysis
 - **LDA** - Latent Dirichlet Allocation - probabilistic model that identifies common topics in docs
 - **BERT** - Google's NLP model that can learn a wide breadth of tasks relating to language
- Learn **data cleaning & processing** techniques
- Revamp the **dashboard** from last semesters - make it interactive and informative for instructors



Metrics/Data We're Using/ Why this Data is Relevant

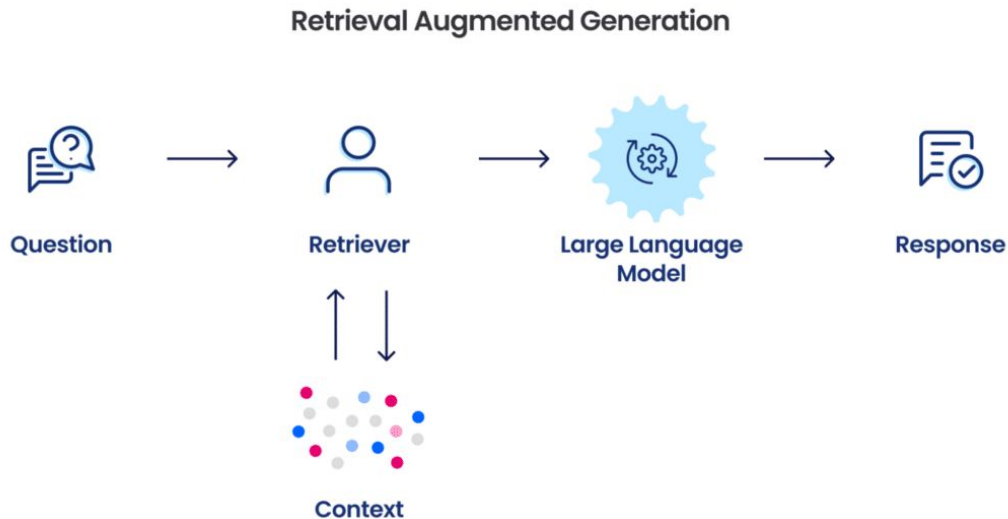
- Text Extraction
 - Pandas - data cleaning
 - LDA - data exploration
 - TF-IDF - term frequency
 - BERT - predict classification



Machine Learning

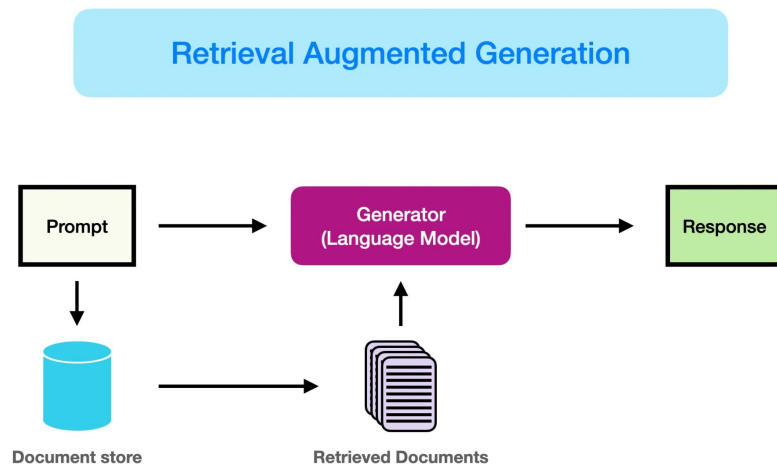
Goals for the Semester

- Research & Implement RAG
 - Techniques to create a synthetic dataset
 - Will be able to improve the output
- Explore a larger variety of models
 - Chat-GPT API, Cohere, and other models
- Refine the prompt given to the model
 - Test variables such as temperature, entropy, and TTR

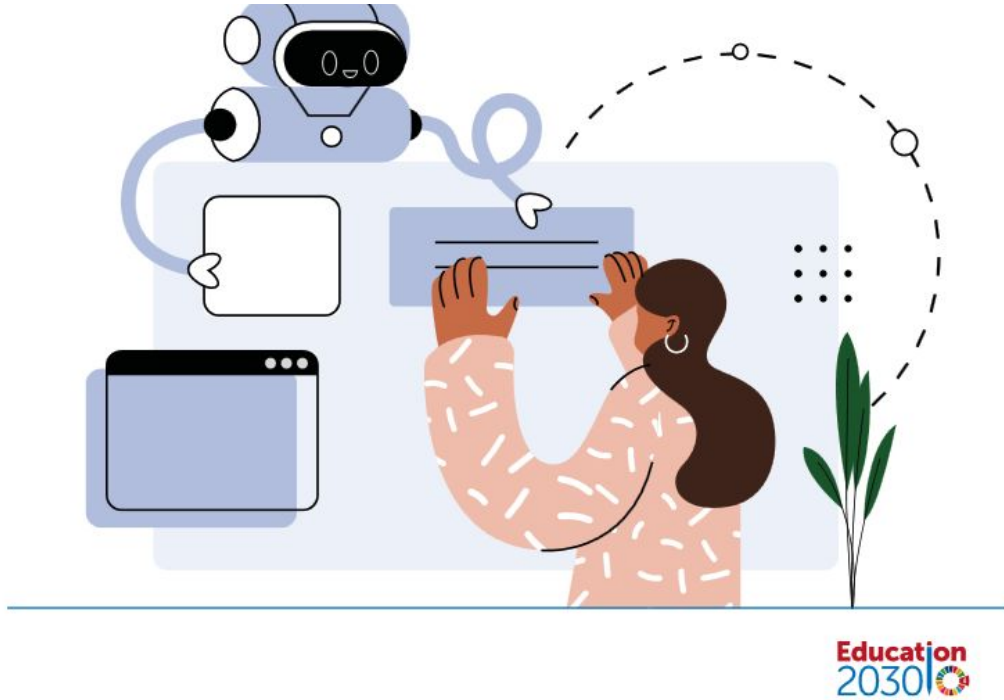


Progress So Far:

- Researched RAG:
 - Retrieval augmented generation
 - Searches external sources prior to generating a response
 - Reduces hallucinations
 - Uses vector search
 - Retrieval based on similarity
 - Has shown to improve outputs



Steps To Take This Semester:



1. Analyze the dataset
2. Explore how to implement RAG and how I can integrate it
3. Reach out to the Harvard Researcher that presented last semester
4. Try to get access to the Chat-GPT API
5. Work with the model to improve outputs to mimic a MOOC dataset

Literature Review

Literature Review:

Goals:

- Understand how NLP can improve online discussion analysis
- Identify existing methods for topic modeling (LDA, TF-IDF) and sentiment analysis
- Explore how discussion forums impact student engagement and learning outcomes
- Find best practices for designing instructor-friendly dashboards
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Body of Knowledge:

- Sentiment analysis with transformer models evaluates discussion tone, providing insights into student engagement and struggles (Hernandez et al., 2024)
- Despite the promising results of NLP in educational research, there are instances where studies fail to effectively integrate educational theory with the needs of stakeholders” (Mello et al., 2024)

Conclusion

- Improve course design
- Provide inclusive learning environments
- Promote student's engagements
- Help educators refine teaching strategies
- Ensure a more responsive and adaptive online learning experience.

Citations

Galán Hernández, J. J., Marín Díaz, G., & Mariscal, G. (2024). Methodology for analyzing educational forums with NLP: Searching for economic terms. In *Teaching innovations in economics* (pp. 77–97). Springer.

https://doi.org/10.1007/978-3-031-72549-4_4

Mello, R. F., Freitas, E., Cabral, L., Pereira, F. D., Rodrigues, L., Rakovic, M., Raniel, J., & Gašević, D. (2024). Words of wisdom: A journey through the realm of natural language processing for learning analytics—A systematic literature review. *Journal of Learning Analytics*, 11(3), 82–105.

<https://doi.org/10.18608/jla.2024.8403>

Thanks for listening!

Q&A